Polymeric Chemistry
Rheology

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1 Theory

In rheology, the flow behaviour of a substance under different forces is studied. This behaviour can be determined using rotational or oscillatory measurements.

1.1 Basic concepts

To measure rheological properties, the sample is placed between two plates, the upper one of which is capable of moving. When this plate is set in motion, the sample experiences a force and moves along with it, but exhibits a velocity gradient from the lower to the upper plate. A visual representation of this is shown in Figure 1.

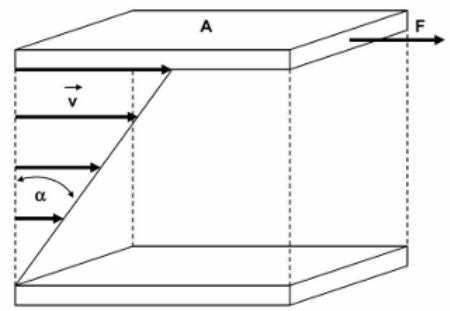

Figure 1: The diagram shows two plates with an area A, the upper one of which exerts a force F on the specimen, causing it to experience a velocity gradient $\vec{v}$ and to move through an angle α. [1]

Key parameters that can be detected or specified in this context include deformation γ, deformation rate $ga\dot{m}ma$, shear stress τ and steady-state shear viscosity η. These parameters are defined by Equations (1–4).

$$\text{Deformation:} \qquad \gamma \equiv \frac{dx}{dy} = \tan\alpha \tag{1}$$

$$\text{Deformation rate:} \qquad \dot{\gamma} \equiv \frac{d\gamma}{dt} \tag{2}$$

$$\text{Shear stress:} \qquad \tau = \frac{F}{A} = \eta \cdot \dot{\gamma} \tag{3}$$

$$\text{Steady-state shear viscosity:} \qquad \eta = \frac{\tau}{\dot{\gamma}} \tag{4}$$

x and y represent the coordinate axes, with x pointing in the direction of the force F. [1]
If a strain rate is specified and the shear stress is thereby detected, the viscosity can also be determined, allowing these parameters to be investigated as a function of the strain rate. Three types of curve behaviour can be detected for the shear stress or viscosity. The parameters may follow a linear trend or deviate from it. If the parameters deviate, they may become either larger or smaller at higher deformation rates. If a sample exhibits a linear relationship, it is referred to as a Newtonian fluid. If the parameters increase at higher deformation rates, it is referred to as a non-Newtonian shear-thickening fluid; conversely, if they decrease, it is referred to as a non-Newtonian shear-thinning fluid. For example, Figure 2 shows the experimental behaviour of a Newtonian fluid, Figure 3 that of a shear-thinning fluid, and Figure 4 that of a shear-thickening fluid. [1]

1.2 Behaviour of matter

1.2.1 Elastic

The elastic behaviour of a polymer can be compared to that of a rubber band or a spring. When a force is applied, causing deformation or shear stress, the polymer responds by undergoing shear or deformation, and returns to its original state once the force is removed. [1]

1.2.2 Viscous

In the case of viscous behaviour, the material deforms continuously when a force is applied and retains its deformation once the force is removed. [1]

1.2.3 Plastic

In the case of plastic behaviour, the material only deforms once a certain stress level is reached; prior to this, it behaves either elastically or shows no deformation. [1]

1.2.4 viscoelastic

Most materials exhibit viscoelastic behaviour, which describes a combination of elastic and viscous behaviour. When a stress is applied, the material will deform elastically and then continue to deform under further shear stress. If this stress is removed, the deformation is partially reversed, but a slight deformation remains. This residual deformation depends on whether the material exhibits more elastic or viscous behaviour.

1.3 Measurement methods

As mentioned at the start, in rheological measurements the sample is placed between two plates and moved through the upper plate. However, in a plate-plate geometry, the rate of deformation of the material depends on the radius. For this reason, the upper plate is often replaced by a slightly conical plate. This conical plate can then be rotated or oscillated. During rotation, the conical plate can be rotated at different speeds, with either the strain rate or the stress being applied. This allows the viscosity, strain rate and shear stress to be determined. During oscillation, the stress or strain is again applied, and the other quantity is thereby determined Furthermore, the cone plate is oscillated only up to a certain deflection and at a specific frequency. This is used to measure the complex viscosity η, the viscoelasticity and, at the same time, the extent to which the material is more viscous or elastic. The entire process is determined by varying the deflection and subsequently selecting the frequency, and follows the equations below.

1.4

2 Results

2.1 Rotational measurements

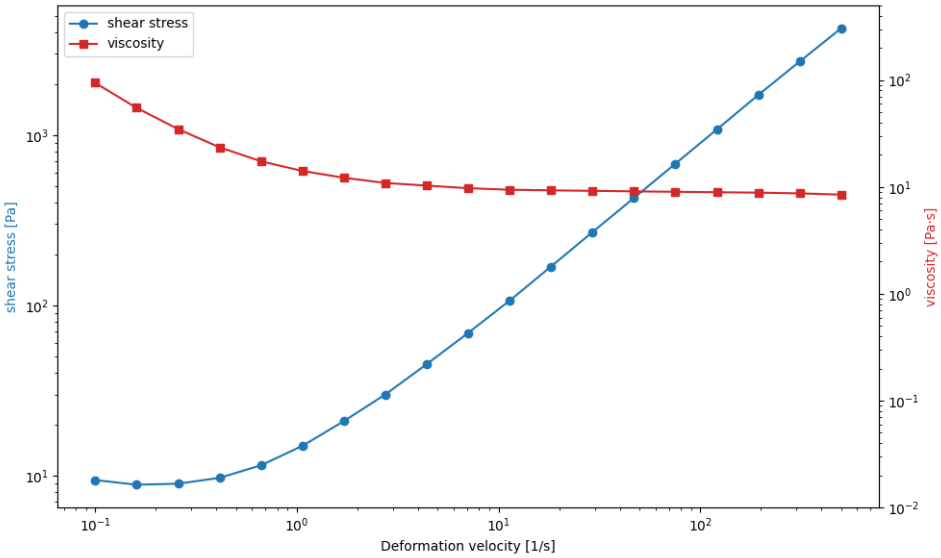

Figure 2: The graph shows the measurements of honey. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

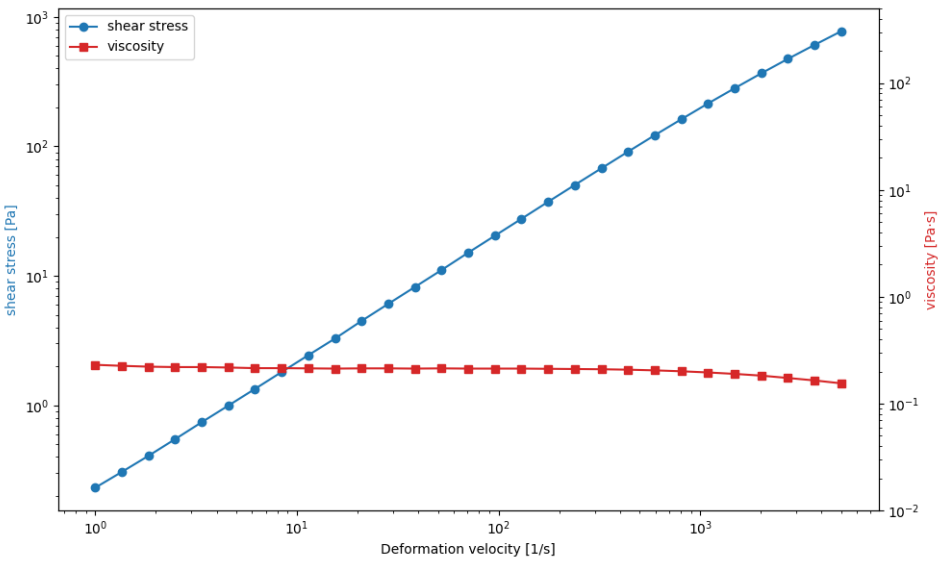

Figure 3: The graph shows the measurements of 14%-PVP solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

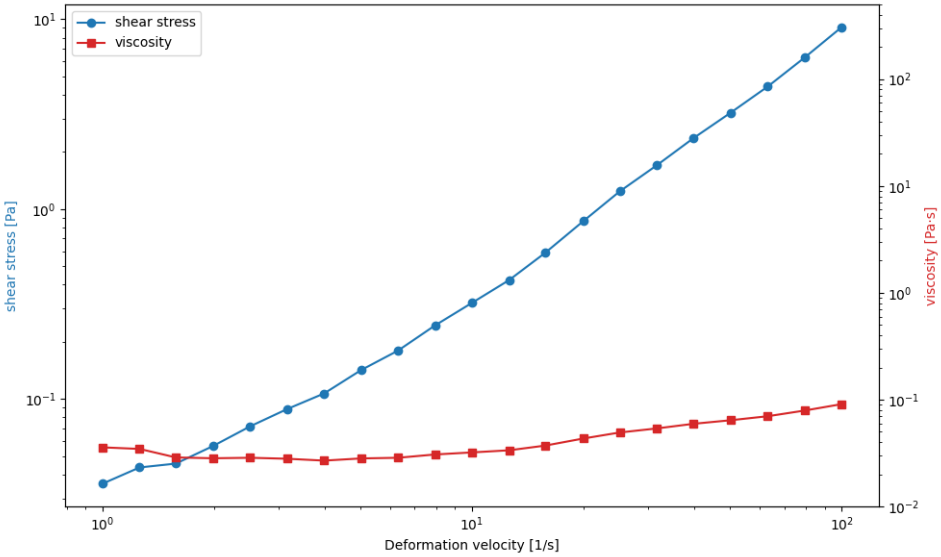

Figure 4: The graph shows the measurements of a starch solution. On the left y-axis the shear stress and on the right y-axis the velocity is plotted, both of these are plotted against the deformation velocity.

2.2 Time dependant measurement of the viscosity

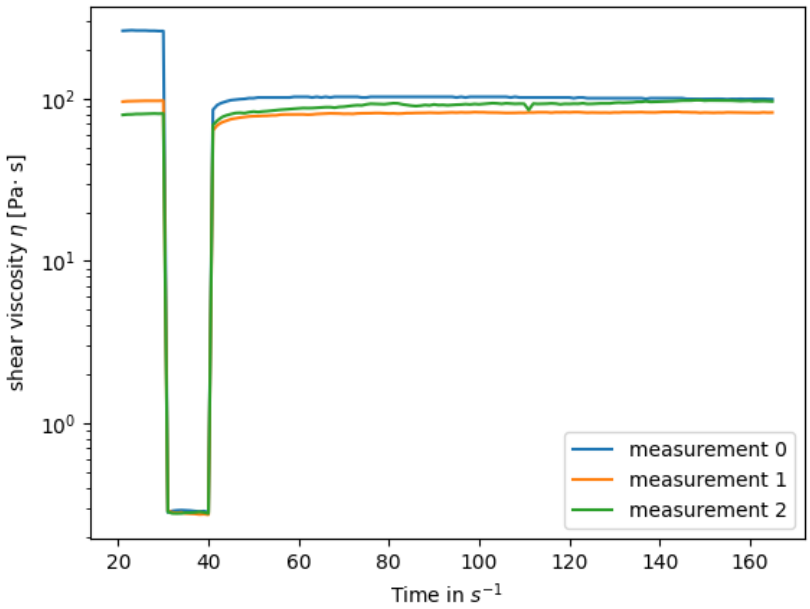

Figure 5: The graph shows the three time dependant viscosity measurements of ketchup. The shear viscosity is plotted against the duration.

2.3 Oszillational measurements to determine the viscoelastic properties

2.3.1 amplitude sweep

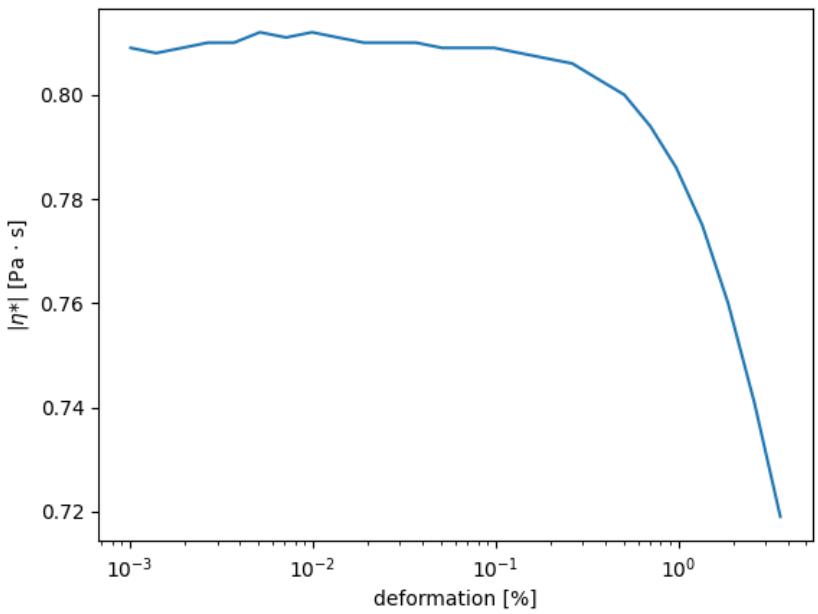

Figure 6: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant frequency.

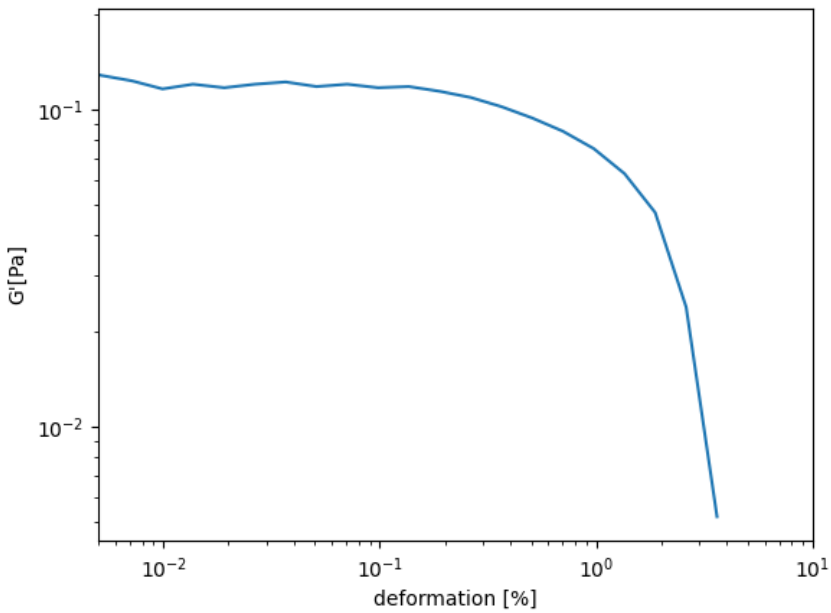

Figure 7: This graph shows the storage module G' plotted against the deformation of 20%-PVP solution at constant frequency.

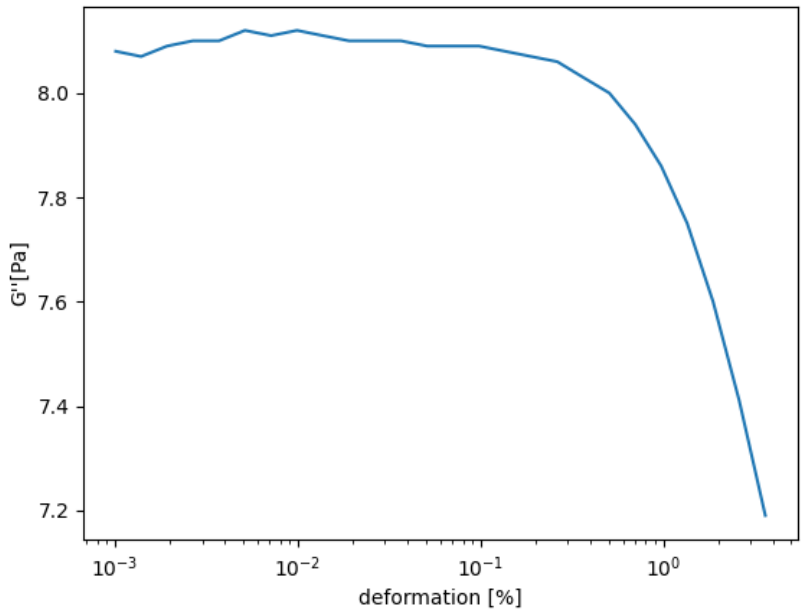

Figure 8: This graph shows the loss module G" plotted against the deformation of 20%-PVP solution at constant frequency.

2.3.2 frequency sweep

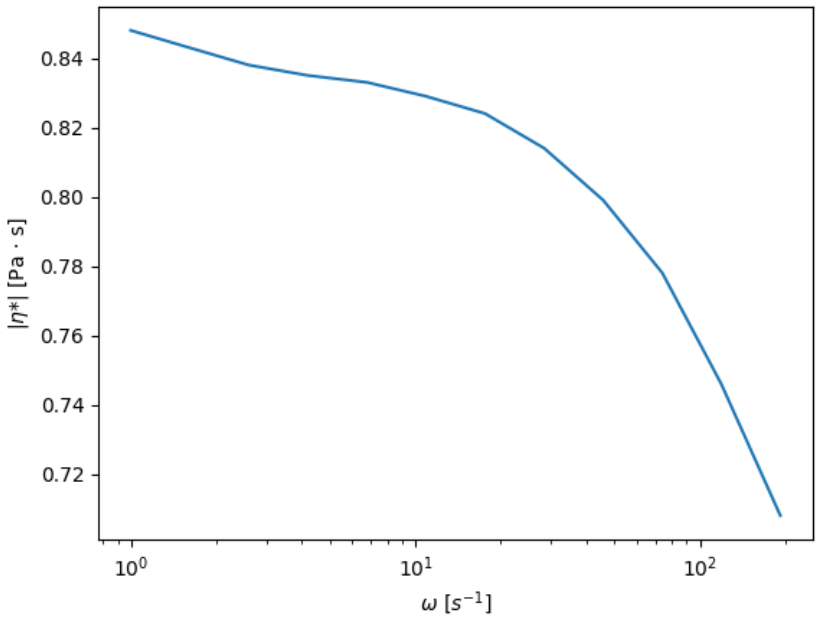

Figure 9: This graph shows the complex viscosity plotted against the deformation of 20%-PVP solution at constant amplitude.

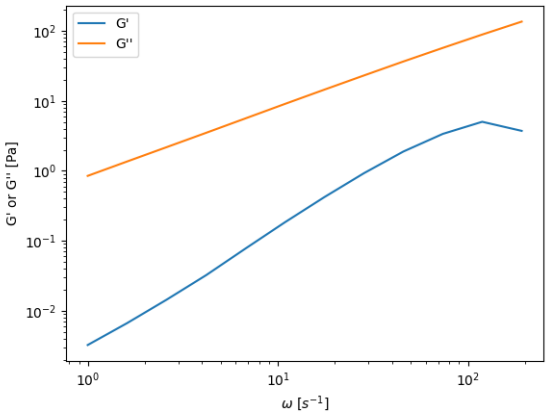

Figure 10: This graph shows the storage module G' loss module G" plotted against the deformation of 20%-PVP solution at constant amplitude.

References

[1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.